```
%%1번문제
A = [1 4 2; 2 4 100; 7 9 7; 3 pi 42];
B = log(A);
a = B(2,:)
```

```
a = 1×3
    0.6931    1.3863    4.6052
```

```
b = sum(B(2,:))
```

```
b =
6.6846
```

```
c = B(:, 2).*A(:,1)
```

```
c = 4×1
    1.3863
    2.7726
   15.3806
    3.4342
```

```
d = max(B(:,2).*A(:,1))
```

```
d =
15.3806
```

```
e = sum(A(1,:) ./ B(1:3, 3)')
```

```
e =
3.3391
```

```
%%2번문제
%%(a)
v0 = 50;
A = 50;
g = 9.81;
t_hit = 2 * v0 * sind(A) / g;%%충돌시간
t1 = v0 * sind(A) / g;%%최고 높이일 때 시간
y_max = v0 * sind(A) * t1 - 0.5 * g * t1^2;
disp(t_hit)
```

```
    7.8088
```

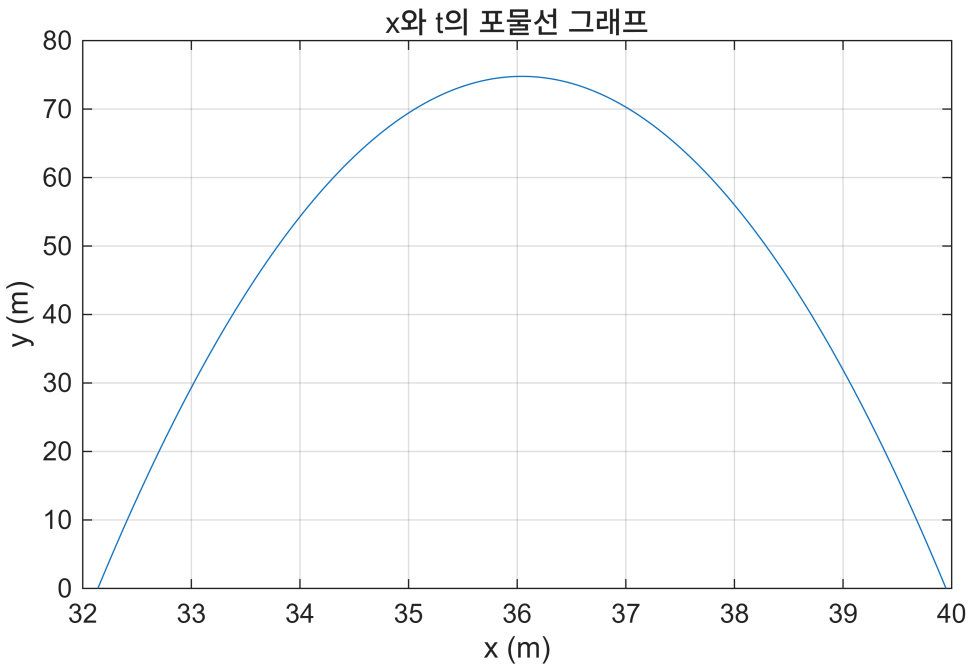
```
disp(y_max)
```

```
   74.7737
```

```
%%(b)
t = linspace(0, t_hit, 1000);
x = v0 * cosd(A) + t;
y = v0 * sind(A) * t - 0.5 * g * t.^2;

figure;
plot(x, y);
xlabel('x (m)');
```

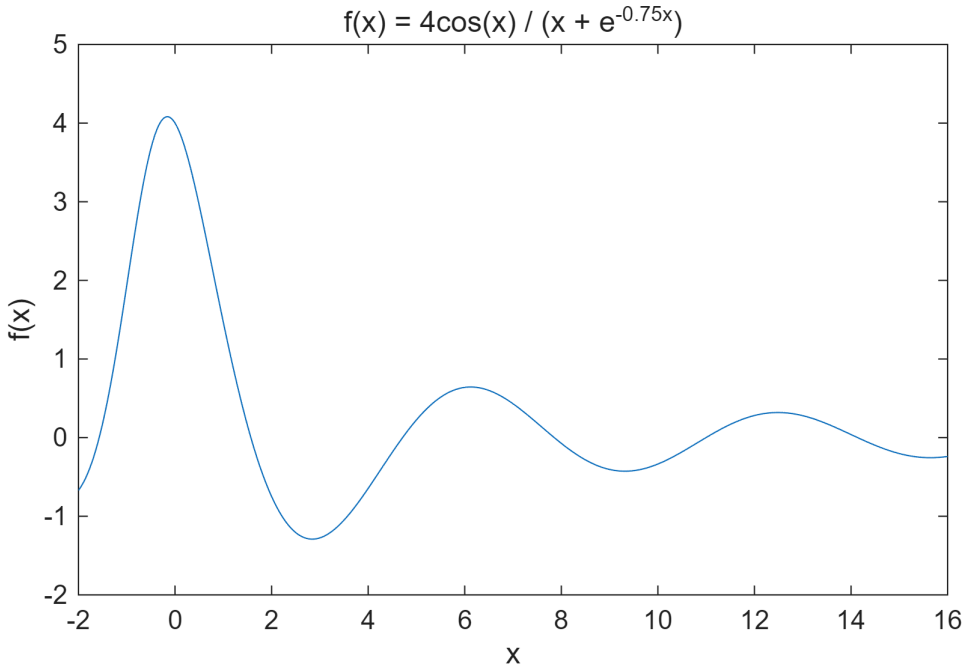
```
ylabel('y (m)');
title('x와 t의 포물선 그래프');
grid on;
```




```
%%3번문제
clear; clc;

x = linspace(-2, 16, 1000);   % 충분히 많은 점
y = 4*cos(x) ./ (x + exp(-0.75*x));

figure;
plot(x, y);
xlabel('x');
ylabel('f(x)');
title('f(x) = 4cos(x) / (x + e^{-0.75x})')
```




```
%%4번문제
clear; clc;

price = [300; 550; 400; 250; 500];
A = [5 4 3; 3 2 4; 6 5 3; 3 5 4; 2 4 3];%%구매량
%%(a)
a = price .* A
```

```
a = 5×3
        1500        1200         900
        1650        1100        2200
        2400        2000        1200
         750        1250        1000
        1000        2000        1500
```

```
%%(b)
b = sum(a, 1)
```

```
b = 1×3
        7300        7550        6800
```

```
%%(c)
c = sum(a, 2)
```

```
c = 5×1
        3600
        4950
        5600
        3000
        4500
```

```
%%(d)
d = sum(a(:))
```

```
d =
21650
```

```
%%5번문제
%%(a)
p1 = [5 0 -3 7 10];
p2 = [4 -7 9 2];
p = conv(p1, p2)
```

```
p = 1×8
    20   -35    33    59   -36   -13   104    20
```

```
%%(b)
num = [5 -7 2 8];
den = [6 3 2];
[q, r] = deconv(num, den)
```

```
q = 1×2
    0.8333   -1.5833
r = 1×4
         0         0    5.0833   11.1667
```

```
%%(c)
n = [36 -8 0 2];
d = [36 -5 4 -3];
x = 5;
y = polyval(n,x)/polyval(d,x)
```

```
y =
0.9795
```

```
%%6번문제
clear n;
clc;
%% (a)
n(1).location = 'Smith St.';
n(1).max = 80;
n(1).year = 1928;
n(1).inspection = 2025;

n(2).location = 'Hope Ave.';
n(2).max = 90;
n(2).year = 1950;
n(2).inspection = 2027;

n(3).location = 'Clark St.';
n(3).max = 85;
n(3).year = 1933;
n(3).inspection = 2024;
```

```
n(4).location = 'North Rd.';
n(4).max = 100;
n(4).year = 1960;
n(4).inspection = 2023;

disp(n)
```

```
다음 필드를 포함한 1×4 struct 배열:

  location
  max
  year
  inspection
```

```
%% (b)
n(3).inspection = 2026;

disp(n(3))
```

```
      location: 'Clark St.'
           max: 85
          year: 1933
    inspection: 2026
```

```
%% (c)
n(5).location = 'Shore Rd.';
n(5).max = 85;
n(5).year = 1997;
n(5).inspection = 2022;

disp(n)
```

```
다음 필드를 포함한 1×5 struct 배열:

  location
  max
  year
  inspection
```